Support-Gated Contrastive Decoding for Reducing Multimodal Hallucination

Anonymous ACL submission

Abstract

Open multimodal language models often produce fluent object and attribute claims that are weakly grounded in the image, and training-free decoding methods are attractive because they can be applied to existing open checkpoints without collecting new supervision. Prior contrastive decoding methods motivate comparing the original image with a distorted visual view, but a global contrastive shift can affect both unsupported hallucinated tokens and visually supported details. We draft Support-Gated Contrastive Decoding (SGCD), a training-free decoder that estimates token-level visual support from original-versus-perturbed visual distributions and applies contrast primarily to likely visual-content tokens with low support. The planned evidence matrix evaluates LLaVA-1.5-7B and Qwen2.5-VL-7B-Instruct on POPE, CHAIR/MSCOCO, HallusionBench, and MME against greedy decoding, nucleus sampling, and VCD. The full matrix is still running, so the headline hallucination change is `TBD_RESULT_ABSTRACT_HALLUCINATION_DELTA`, the VQA/perception guardrail is `TBD_RESULT_ABSTRACT_GUARDRAIL_DELTA`, and the final implication remains `PLACEHOLDER_ABSTRACT_IMPLICATION`.

1 Introduction

Open multimodal language models (MLLMs) have become a practical interface for visual question answering, captioning, and multimodal assistance, but their answers can include objects or attributes that are plausible under the language context and absent from the image. This failure mode is well established by object-hallucination and caption-hallucination evaluations such as POPE and CHAIR, and by broader multimodal diagnostics such as HallusionBench and MME (Li et al., 2023b; Rohrbach et al., 2018; Guan et al., 2023; Fu et al., 2023). The resulting paper problem is not only whether hallucination exists; it is whether a decoder can suppress unsupported visual-content tokens without making the model terse, evasive, or worse at general visual perception.

We define a hallucinated visual claim as a generated object, attribute, relation, or visual yes/no assertion whose support is weaker in the image than in the language prior implied by the prompt and prefix. This definition deliberately excludes stylistic verbosity and non-visual fluency errors, because a decoding method that simply shortens every answer can appear to reduce hallucination while losing useful visual detail. It also excludes oracle access to benchmark annotations: at inference time the decoder sees only the image, prompt, model state, and its own competing next-token distributions. The central design constraint is therefore selective intervention. A useful training-free decoder should know when a next-token decision is visual-content bearing, estimate whether that token is image-supported, and leave ordinary function words or strongly supported content largely unchanged.

Training-free methods are a natural fit for this setting because many strong open MLLMs are already released as fixed checkpoints. Nucleus sampling is a standard status-quo decoding baseline (Holtzman et al., 2019), while Visual Contrastive Decoding (VCD) contrasts logits from the original image and a distorted image to reduce hallucination without model updates (Leng et al., 2023). OPERA proposes another training-free route through over-trust penalties and retrospective allocation (Huang et al., 2023). These methods motivate a shared question: should the contrastive or penalty signal be global, or should it focus on the tokens that appear visually unsupported?

This draft studies the second option. Support-Gated Contrastive Decoding (SGCD) estimates a visual-support margin for candidate tokens by comparing the evaluated MLLM's next-token distribution under the original image with the distribution under a deterministic visual perturbation. Instead of treating every token equally, SGCD applies its low-support penalty mainly to lexical tokens likely to express visual content. The mechanism uses no detector, no gold object list, no benchmark label, and no prompt change relative to the baselines. This makes the method a decoding-time intervention rather than a new model, reranker, or external fact-checking pipeline.

The intended contribution is threefold. First, SGCD defines a non-oracle, token-level support gate for training-free multimodal decoding. Second, the evaluation plan compares SGCD against greedy, nucleus, and VCD on a four-family public benchmark matrix with two open 7B-class MLLM backbones. Third, the analysis plan tests whether any future hallucination reduction comes from the support gate rather than from answer shortening or abstention. The outcome-dependent form of these contributions is intentionally unresolved: `TBD_RESULT_INTRO_MAIN_CLAIM`, `TBD_RESULT_INTRO_BASELINE_COMPARISON`, and `TBD_RESULT_INTRO_GUARDRAIL`.

2 Related Work

Hallucination benchmarks. CHAIR introduced object-hallucination measurement for image captioning by comparing generated object mentions to image annotations (Rohrbach et al., 2018). POPE reframed object hallucination as a yes/no probing problem with random, popular, and adversarial negative objects (Li et al., 2023b). HallusionBench stresses entangled visual illusion and language hallucination, while MME provides a broader perception and cognition suite that can serve as a guardrail against degrading general visual ability (Guan et al., 2023; Fu et al., 2023). This paper uses these benchmark families because they test different hallucination surfaces rather than variants of one local diagnostic.

Open MLLM backbones. The target backbones come from two open model families. LLaVA and LLaVA-1.5 are common baselines for visual instruction tuning and hallucination studies (Liu et al., 2023b,a). Qwen-VL, Qwen2-VL, and Qwen2.5-VL provide a second current MLLM family with public checkpoints and stronger perception coverage (Bai et al., 2023; Wang et al., 2024; Bai et al., 2025). Other open multimodal systems such as BLIP-2, InstructBLIP, and MiniGPT-4 help contextualize the broader architecture landscape (Li et al., 2023a; Dai et al., 2023; Zhu et al., 2023).

Training-free hallucination mitigation. VCD is the required strong baseline because it directly contrasts original and visually distorted branches at decoding time (Leng et al., 2023). OPERA addresses a related failure through an over-trust penalty and retrospection allocation (Huang et al., 2023). Woodpecker shows that external correction pipelines can reduce hallucination, but it is outside this paper's pure decoding-time scope because it relies on additional correction machinery (Yin et al., 2023). SGCD is positioned between status-quo decoding and global contrastive decoding: it keeps VCD's training-free original-versus-perturbed signal, but gates the intervention by token-level support and lexical visual-content status.

3 Method

3.1 Problem Setting

Given an image x, a text prompt q, and a generated prefix $y_{<t}$, the decoder must select the next token without access to benchmark labels, image annotations, detector outputs, or an external correction model. All methods share the same prompt and answer budget within each benchmark. SGCD differs only in how it transforms next-token scores during decoding. Figure 1 summarizes this training-free flow and highlights that the fixed model state is unchanged while candidate token scores are gated. This shared-interface constraint is important for attribution: a future improvement cannot be credited to a rewritten prompt, extra visual parser, or post-hoc answer filter.

3.2 Visual-Support Margin

At decoding step t, SGCD runs the evaluated MLLM on the original image and on a deterministic perturbed image. Let z_i^+ and z_i^- be

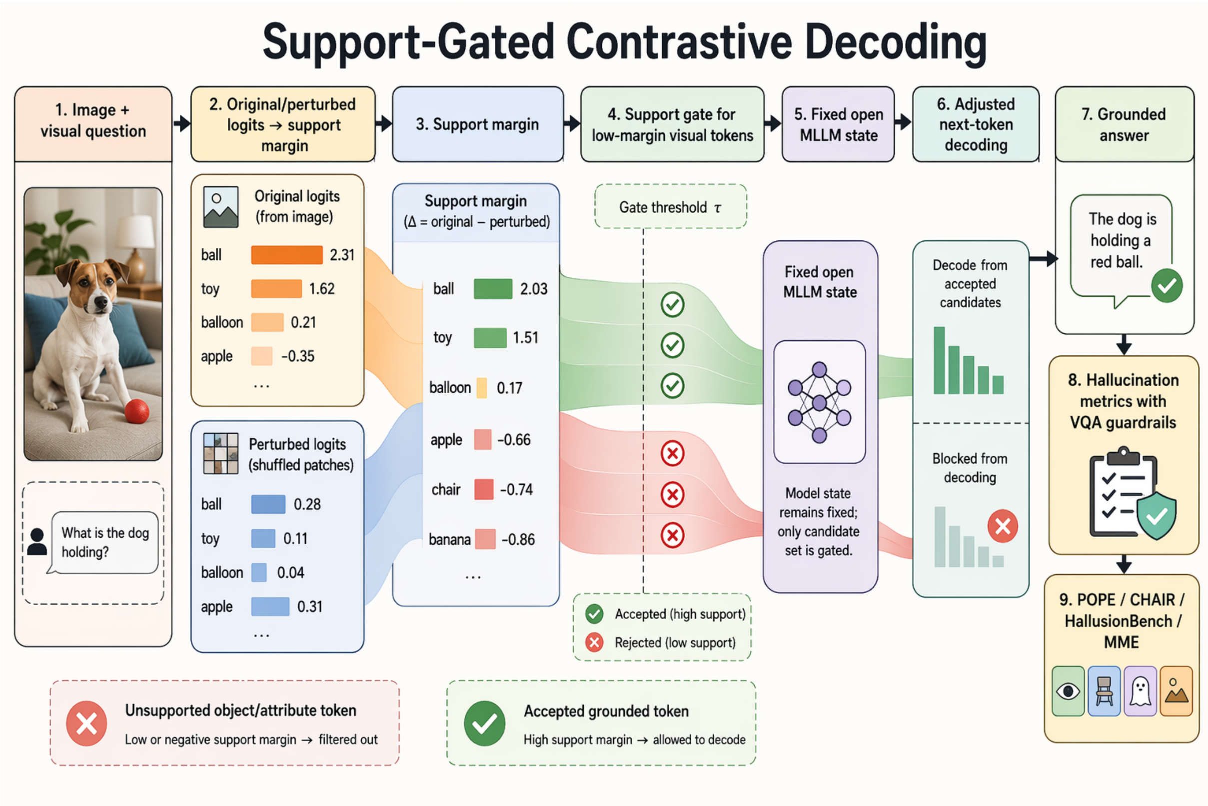

Figure 1: Support-Gated Contrastive Decoding overview. SGCD compares original and perturbed visual branches from the same fixed open MLLM, estimates token-level support margins, gates low-support visual-content candidates, and decodes from adjusted next-token scores. The bottom strip names the planned benchmark evidence families; numerical outcome claims remain deferred to completed full-matrix analysis.

the original and perturbed logits for token i, and let ℓ_i^+ and ℓ_i^- be the corresponding log probabilities. The visual-support margin is

$$m_i = \ell_i^+ - \ell_i^-.$$

Tokens with larger positive margins are more supported by the original visual evidence relative to the perturbed branch. The candidate set is the union of a top-k set and a nucleus set under the original distribution, matching the plan-stage SGCD contract.

The perturbation branch is not intended to be a second view of the correct answer. It is a contrastive probe of which next tokens depend on the image representation. If a token remains equally likely after the visual signal is weakened, SGCD treats it as less visually supported at that step. If the token becomes less likely under perturbation, the positive margin is evidence that the original image is contributing to the decision. This interpretation is local to the next-token distribution; SGCD does not claim to identify complete objects, verify a whole sentence, or recover a scene graph.

3.3 Support-Gated Contrast

For each candidate token, SGCD computes a low-support deficit

$$d_i = \max(0, \tau_s - m_i)$$

and a deterministic lexical content indicator c_i. The content indicator normalizes the token string, removes punctuation and common function words, and marks likely object or attribute tokens without reading the image, benchmark label, or any detector output. The adjusted next-token score is

$$\tilde{z}_i = z_i^+ + \alpha \cdot \mathrm{clip}(m_i, -\gamma, \gamma) - \lambda c_i d_i.$$

Greedy decoding selects the maximum original score, VCD uses the contrastive branch without the support gate, and SGCD applies the support-gated penalty. The ablations remove the content gate or the support gate to test whether the selective mechanism matters.

The scoring rule separates three effects. The clipped contrast term preserves the intuition of contrastive decoding: tokens that depend on the original image can receive a controlled boost relative to the perturbed branch. The deficit term activates only below the support threshold τ_s, so high-margin visual tokens are not penalized merely because they are nouns or attributes. The content indicator c_i limits the penalty to likely visual-content tokens, reducing the chance that the method globally distorts grammar, answer format, or benchmark-specific short responses. The temperature, nucleus, and answer-length settings are held fixed within a benchmark condition; SGCD's added compute is the perturbed forward pass and candidate-level score transform.

3.4 Decoding Policy

At each step, SGCD first obtains the original distribution and constructs a bounded candidate set. It then obtains the perturbed distribution for the same prompt prefix, computes margins for the candidates, applies the support-gated score transform, and samples or selects according to the condition's decoding policy. For greedy SGCD, the final decision is the maximum adjusted score. For stochastic SGCD variants, the adjusted scores are renormalized inside the same nucleus budget used by the matched nucleus baseline. End-of-sequence handling and maximum-answer budgets are inherited from the benchmark harness so that an apparent hallucination reduction cannot come from silently changing stopping rules.

3.5 Guardrails Against Trivial Suppression

A decoding penalty can look helpful for the wrong reason if it suppresses all content-bearing words, increases abstentions, or collapses captions to short generic answers. SGCD therefore has three built-in guardrails that are independent of the eventual numerical results. First, the support deficit is margin-based rather than a blanket noun penalty. Second, the content gate affects only candidate tokens and never deletes an already generated span. Third, the planned analysis includes length, yes-ratio, abstention, and MME perception checks before any positive claim is allowed. If those checks fail, the final paper must report SGCD as a boundary case or negative result rather than a hallucination improvement.

3.6 Non-Oracle Constraints

SGCD is designed to be auditable as a training-free decoder. It does not tune parameters on full-test outcomes, does not change prompts, and does not use object lists, answer keys, or external visual detectors during inference. Its only visual signal is the same model's distributional response to the original image and a label-free perturbation. The future analysis artifacts must preserve these constraints before any result claim is backfilled. Any condition that uses benchmark labels, detector output, or manual answer correction at decoding time should be treated as an oracle diagnostic rather than the proposed method.

4 Experimental Setup

Models. The planned full matrix evaluates LLaVA-1.5-7B and Qwen2.5-VL-7B-Instruct. These model names are paper-facing checkpoint families; machine-specific execution details are excluded from the manuscript body.

Benchmarks. The benchmark families are POPE, CHAIR/MSCOCO, HallusionBench, and MME. The benchmark provenance artifact records public source URLs, task counts, hashes, and scorer wrappers for each family. POPE tests object-existence decisions, CHAIR tests caption object hallucination, HallusionBench tests image-context conflicts, and MME is used as a broader perception guardrail. The matrix intentionally mixes yes/no probes, free-form caption scoring, adversarial hallucination diagnostics, and broad perception questions. This prevents the paper from treating one benchmark family as a proxy for all multimodal hallucination behavior.

Baselines and method conditions. The required conditions are greedy decoding, nucleus sampling, VCD, and SGCD. Greedy and nucleus are status-quo decoding controls; VCD is the required training-free contrastive baseline; SGCD is the proposed support-gated variant. The planned ablations are SGCD without content gating and SGCD without support gating. All conditions are evaluated through the same benchmark harness, answer budgets, and scoring interfaces. The baselines differ only in the decoding-time score policy: greedy uses the original branch, nucleus samples from the original branch, VCD applies ungated original-versus-perturbed contrast, and SGCD applies the support gate described above.

Metrics and guardrails. The paper-facing hallucination metrics will be benchmark-specific: POPE accuracy/F1 and yes-ratio diagnostics, CHAIR sentence/image hallucination rates and length controls, HallusionBench accuracy-style group scores, and MME perception/cognition guardrails. Paired significance will use McNemar-style tests for paired binary outcomes and bootstrap intervals for CHAIR-style caption metrics after canonical analysis tables exist. The exact result tokens are listed in the project placeholder ledger and cannot be replaced until their owning source artifacts exist.

Run and backfill policy. The current full matrix is live rather than terminal. Therefore this draft treats the Results, Analysis, and Failure Cases sections as scaffolds and contains no final outcome claims. Backfilling is allowed only when completed scored rows cover the relevant model, benchmark, and method condition, and when a canonical analysis artifact regenerates the corresponding table entry. Live partial rows, terminal logs without scored rows, and manually read progress counters are not evidence for paper claims.

Seed and stopping policy. The deterministic decoding conditions use fixed benchmark prompts and fixed answer budgets. Stochastic conditions will report the sampling settings and seed policy used by the harness. Runs may be resumed by the experiment controller, but duplicate rows for the same semantic task and method must be de-duplicated in analysis before any aggregate is reported. If a condition stops early because of model loading, prompt length, or evaluator configuration, that condition is a run-stage blocker rather than a negative result for SGCD.

5 Results

Table 1 is a placeholder scaffold for the main cross-benchmark matrix. It is intentionally not a result table yet: every outcome cell must be backfilled from future raw rows and analysis tables, not from live partial progress.

The table-level backfill token is `PLACEHOLDER_MAIN_RESULTS_TABLE`.

The paper will report SGCD as better than a baseline only if the completed analysis supports it. If the analysis shows improvement only on one family, degradation on MME, length collapse, or non-significant paired differences, the final text must soften or reject the positive thesis rather than filling this scaffold with selective numbers. The expected reader-facing comparison order is greedy and nucleus first, VCD second, and SGCD last. This order keeps the proposed method anchored against both common decoding practice and the closest training-free contrastive baseline.

6 Analysis and Ablation

The analysis section will test the mechanism rather than only reporting aggregate scores. The support-margin diagnostic will check whether low visual-support margins are enriched among hallucinated object or attribute decisions: `TBD_RESULT_SUPPORT_MARGIN_SEPARATION`. The support-gate ablation will test whether removing the support gate weakens SGCD: `TBD_RESULT_ABLATION_SUPPORT_GATE_EFFECT`. The content-gate ablation will test whether lexical gating helps avoid broad suppression of non-visual tokens: `TBD_RESULT_ABLATION_CONTENT_GATE_EFFECT`. These ablations are necessary because a global contrastive method and a support-gated method can both change the next-token distribution while producing very different failure profiles. If the no-support-gate ablation matches SGCD, the support margin is not doing useful work. If the no-content-gate ablation improves hallucination while harming length or perception guardrails, the final paper should frame lexical gating as a safety valve rather than a pure accuracy booster.

Guardrail analyses are equally important. The CHAIR length ratio will be `TBD_RESULT_LENGTH_RATIO`, the abstention shift will be `TBD_RESULT_ABSTENTION_DELTA`, and paired significance support will be `TBD_RESULT_PAIRED_SIGNIFICANCE_SUMMARY`. No final claim about mechanism, significance, or robustness is made in this draft. The qualitative analysis will also inspect cases where SGCD and VCD disagree. Those

Benchmark	Model family	Hallucination outcome	Guardrail	Metric family
POPE	LLaVA / Qwen2.5-VL	TBD_RESULT_POPE_SGCD_VCD_DELTA	TBD_RESULT_POPE_YES_RATIO	paired yes/no object probes
CHAIR/ MSCOCO	LLaVA / Qwen2.5-VL	TBD_RESULT_CHAIR_CHAIRS_DELTA	TBD_RESULT_LENGTH_RATIO	caption object hallucination
HallusionBench	LLaVA / Qwen2.5-VL	TBD_RESULT_HALLUSION_ACC_DELTA	TBD_RESULT_ABSTENTION_DELTA	conflict-sensitive VQA groups
MME	LLaVA / Qwen2.5-VL	TBD_RESULT_MME_HALLUCINATION_SUBTASK	TBD_RESULT_MME_GUARDRAIL_DELTA	perception/cognition guardrail
Aggregate	Both families	TBD_RESULT_MAIN_AGGREGATE_DIRECTION	TBD_RESULT_MAIN_QWEN_DIRECTION	cross-family consistency

Table 1: Main result scaffold. The numerical headline is intentionally deferred until completed full-matrix scored rows and canonical analysis tables exist.

disagreements are the most informative examples because they isolate the effect of selective support gating from the shared original-versus-perturbed contrast.

7 Failure Cases

Final failure cases must be selected from saved scored rows after the full matrix and qualitative selection manifest exist. The expected taxonomy includes visually ambiguous objects, attribute confusions, language-prior yes bias, over-penalized true visual detail, and perturbation-insensitive prompts. The first paper-facing example is `PLACEHOLDER_FAILURE_CASE_VISUAL_AMBIGUITY`; the second is `PLACEHOLDER_FAILURE_CASE_OVER_PENALTY`; and any Qwen-specific pattern is `PLACEHOLDER_FAILURE_CASE_QWEN_PATTERN`. If saved rows do not support a category, that category must be removed. Failure cases will be reported as paired method outputs for the same image and prompt, not as isolated anecdotes. Each selected example must include the benchmark family, the compared methods, the relevant ground-truth or scorer signal, and why SGCD helped, failed, or traded one error type for another.

8 Conclusion

This draft proposes SGCD as a training-free, support-gated contrastive decoder for open MLLMs. The method is fully specified, and the benchmark matrix is planned, but the empirical conclusion is still pending: `TBD_RESULT_CONCLUSION_MAIN` and `TBD_RESULT_CONCLUSION_SCOPE`. The final paper must either backfill these placeholders from completed full-scale evidence or revise the thesis.

Limitations

This is a bounded overlap draft while the run stage remains active. The latest ground-truth record shows no completed SGCD or Qwen full-matrix evidence, so the draft is not a final EMNLP-ready paper. The method is also limited to training-free decoding with open MLLMs where original and perturbed visual branches can be evaluated. If future evidence shows that SGCD reduces hallucination mainly by shortening answers or increasing abstention, the positive thesis should be rejected or narrowed.

Ethical Considerations

Hallucination reduction in MLLMs matters for assistive and high-stakes visual settings, but a decoding method should not be represented as guaranteeing visual truthfulness. The final paper should report failures and guardrails clearly, avoid overstating safety, and release code and evaluation artifacts with enough provenance for independent replication. No benchmark result should be used to imply reliability outside the evaluated public datasets and model families.

References

Jinze Bai, Shuai Bai, Shusheng Yang, Shijie Wang, Sinan Tan, Peng Wang, Junyang Lin, Chang Zhou, and Jingren Zhou. 2023. Qwen-vl: A versatile vision-language model for understanding, localization, text reading, and beyond. *Preprint*, arXiv:2308.12966.

Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibo Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, Humen Zhong, Yuanzhi Zhu, Mingkun Yang, Zhaohai Li, Jianqiang Wan, Pengfei Wang, Wei Ding, Zheren Fu, Yiheng Xu, and 8 others. 2025. Qwen2.5-vl technical report. *Preprint*, arXiv:2502.13923.

Wenliang Dai, Junnan Li, Dongxu Li, Anthony Meng Huat Tiong, Junqi Zhao, Weisheng Wang, Boyang Li, Pascale Fung, and Steven Hoi. 2023. Instructblip: Towards general-purpose vision-language models with instruction tuning. *Preprint*, arXiv:2305.06500.

Chaoyou Fu, Peixian Chen, Yunhang Shen, Yulei Qin, Mengdan Zhang, Xu Lin, Jinrui Yang, Xiawu Zheng, Ke Li, Xing Sun, Yunsheng Wu, Rongrong Ji, Caifeng Shan, and Ran He. 2023. Mme: A comprehensive evaluation benchmark for multimodal large language models. *Preprint*, arXiv:2306.13394.

Tianrui Guan, Fuxiao Liu, Xiyang Wu, Ruiqi Xian, Zongxia Li, Xiaoyu Liu, Xijun Wang, Lichang Chen, Furong Huang, Yaser Yacoob, Dinesh Manocha, and Tianyi Zhou. 2023. Hallusionbench: An advanced diagnostic suite for entangled language hallucination and visual illusion in large vision-language models. *Preprint*, arXiv:2310.14566.

Ari Holtzman, Jan Buys, Li Du, Maxwell Forbes, and Yejin Choi. 2019. The curious case of neural text degeneration. *Preprint*, arXiv:1904.09751.

Qidong Huang, Xiaoyi Dong, Pan Zhang, Bin Wang, Conghui He, Jiaqi Wang, Dahua Lin, Weiming Zhang, and Nenghai Yu. 2023. Opera: Alleviating hallucination in multi-modal large language models via over-trust penalty and retrospection-allocation. *Preprint*, arXiv:2311.17911.

Sicong Leng, Hang Zhang, Guanzheng Chen, Xin Li, Shijian Lu, Chunyan Miao, and Lidong Bing. 2023. Mitigating object hallucinations in large vision-language models through visual contrastive decoding. *Preprint*, arXiv:2311.16922.

Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. 2023a. Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. *Preprint*, arXiv:2301.12597.

Yifan Li, Yifan Du, Kun Zhou, Jinpeng Wang, Wayne Xin Zhao, and Ji-Rong Wen. 2023b. Evaluating object hallucination in large vision-language models. *Preprint*, arXiv:2305.10355.

Haotian Liu, Chunyuan Li, Yuheng Li, and Yong Jae Lee. 2023a. Improved baselines with visual instruction tuning. *Preprint*, arXiv:2310.03744.

Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. 2023b. Visual instruction tuning. *Preprint*, arXiv:2304.08485.

Anna Rohrbach, Lisa Anne Hendricks, Kaylee Burns, Trevor Darrell, and Kate Saenko. 2018. Object hallucination in image captioning. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 4035–4045, Brussels, Belgium. Association for Computational Linguistics.

Peng Wang, Shuai Bai, Sinan Tan, Shijie Wang, Zhihao Fan, Jinze Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Yang Fan, Kai Dang, Mengfei Du, Xuancheng Ren, Rui Men, Dayiheng Liu, Chang Zhou, Jingren Zhou, and Junyang Lin. 2024. Qwen2-vl: Enhancing vision-language model's perception of the world at any resolution. *Preprint*, arXiv:2409.12191.

Shukang Yin, Chaoyou Fu, Sirui Zhao, Tong Xu, Hao Wang, Dianbo Sui, Yunhang Shen, Ke Li, Xing Sun, and Enhong Chen. 2023. Woodpecker: Hallucination correction for multimodal large language models. *Preprint*, arXiv:2310.16045.

Deyao Zhu, Jun Chen, Xiaoqian Shen, Xiang Li, and Mohamed Elhoseiny. 2023. Minigpt-4: Enhancing vision-language understanding with advanced large language models. *Preprint*, arXiv:2304.10592.

A Reproducibility Appendix

This appendix will be expanded after the full run and analysis finish. The planned methods are greedy, nucleus, VCD, and SGCD; the planned benchmarks are POPE, CHAIR/MSCOCO, HallusionBench, and MME; the planned backbones are LLaVA-1.5-7B and Qwen2.5-VL-7B-Instruct.

A.1 Replay Interface

The final supplementary package will provide a neutral replay interface with named targets for benchmark preparation, model-specific inference, analysis aggregation, and paper-table regeneration. The interface will expose paper-facing method names and benchmark families rather than local runner paths or workstation configuration. Each target will declare the evaluated checkpoint family, decoding policy, prompt template version, answer budget, sampling settings when applicable, and expected output schema. Hardware-specific launch details and storage locations will remain in private run manifests and logs, not in the paper text.

A.2 Artifact Inventory

The final artifact bundle should contain benchmark provenance records, raw scored rows, percondition status files, analysis tables, paired-significance outputs, qualitative-case manifests, and the placeholder ledger used during drafting. The raw-row schema must include benchmark family, semantic task identifier, model family, method condition, decoded answer, scorer output, and any guardrail fields required for length, abstention, or yes-ratio diagnostics. Rows generated by smoke tests or failed launch probes must be marked as non-paper evidence and excluded from main tables.

A.3 Seed and Resume Notes

The deterministic conditions are replayable from the benchmark split, model family, prompt template, and decoding budget. Stochastic conditions require their recorded sampling seed or seed list. If a run is interrupted and resumed, the analysis layer must reconcile duplicate semantic task identifiers before aggregation. Any missing model-method-benchmark condition remains a blocker for final claims and must be represented as an evidence gap rather than silently omitted.

A.4 Placeholder Backfill Contract

Every `TBD_RESULT_*` and `PLACEHOLDER_*` token in the manuscript is listed in the placeholder ledger. Backfilling requires completed raw rows for the owning condition plus a canonical analysis artifact such as a regenerated results table, paired-significance table, guardrail table, or saved-row qualitative manifest. Manual replacement from live progress counters is not allowed.